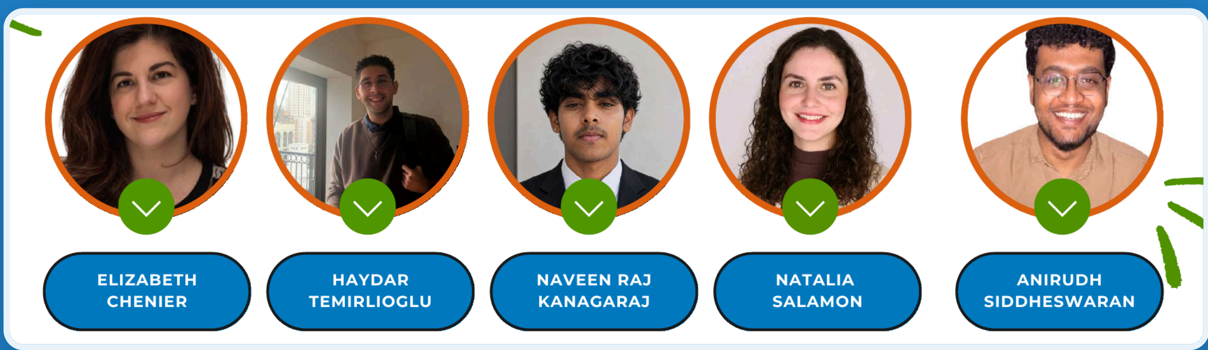
ANALYTICAL TOOLS FOR MARKETERS · MKT 534

Happy Tastes Good Dairy-Free

A Perceptual Mapping and Conjoint Analysis of the Dairy-Free Frozen Dessert Opportunity for Dairy Queen



Prepared for Dr. Zafar Iqbal

Course MKT 534, Analytical Tools for Marketers

Date June 2026

Team Elizabeth Chenier, Naveen Raj Kanagaraj, Natalia Salamon, Anirudh Siddheswaran, Haydar Temirlioglu

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Executive Summary

Dairy Queen should launch a **dairy-free Blizzard at a $4.99 entry price on an oat-milk base**, distributed through its existing 4,200+ walk-in locations. This is the single highest-leverage move available to the brand, and it is supported directly by both quantitative studies in this report.

Two analytical methods point to the same conclusion. **Perceptual mapping** (n = 52, 5 brands across 8 attributes) shows DQ currently sits in the weak "Melting Away / Who?" quadrant, under-differentiated and low on consideration relative to Jeni's, Ben & Jerry's, and Baskin-Robbins. **Conjoint analysis** (n = 50, 7 features across 3 levels, L18 orthogonal design) shows the exact combination of attributes that moves a buyer: low price, indulgent flavor, oat-milk base, and walk-in availability. When DQ's strongest available configuration is modeled, its total preference utility (1.065) **exceeds every competitor** in the set, including Andy's (0.828) and Jeni's (0.856).

The market is large and growing. Roughly **36% of Americans are lactose intolerant** and **41% actively avoid or limit dairy**, while the dairy-free ice cream category is a **$2.65B market growing at 15.3% CAGR**. DQ, meanwhile, has been losing ground, with 42 store closures and flat average unit volume (about $1.165M). A dairy-free line is both a defensive and an offensive play.

Bottom line: the data does not ask DQ whether to enter dairy-free, it tells DQ exactly *how*. Price at $4.99, lead with indulgence, build on oat milk, and sell it where DQ already wins, the walk-in.

Key Findings

- **DQ is poorly positioned today.** On the perceptual map, DQ clusters with Jeni's near the origin but skews toward the low-consideration "Melting Away / Who?" region. Competitors own the high-affinity "Obsessed / Scooped Up" space.
- **Price is the number one purchase driver.** Feature-importance decomposition ranks Price highest at **21.97%**, followed by Availability (20.62%) and Type of Ice Cream (19.41%). Certification and social-cause messaging barely move buyers (6.87% and 1.35%).
- **The winning product is specific, not vague.** Top part-worth utilities: **$4.99 price (0.221)**, **indulgent flavor (0.183)**, **oat-milk base (0.183)**, **walk-in purchase (0.166)**. These four levers define the optimal SKU.
- **A correctly configured DQ wins the category.** Modeled total preference utility: **DQ optimal = 1.065**, beating Andy's (0.828), Jeni's (0.856), Ben & Jerry's (0.796), and Baskin-Robbins (0.743).
- **Demand intent is real.** 91.8% of respondents say environmental, health, or ethical factors influence food choices at least occasionally; 80% would switch to or consider dairy-free; about 55% would pay the same or only $1 more.

1. Objective

The brief was to determine whether, and how, Dairy Queen should enter the dairy-free frozen dessert category, using two complementary marketing-analytics methods rather than intuition.

Research questions:

1. Where does DQ sit in consumers' minds relative to direct competitors? *(Perceptual mapping)*
2. Which product attributes actually drive purchase, and at what levels? *(Conjoint analysis)*
3. Given the above, what is the optimal dairy-free SKU and go-to-market configuration for DQ?

Why now. The category fundamentals are unambiguous:

Signal	Value	Implication
Lactose intolerant (U.S.)	about 36% (roughly 120M people)	Large unaddressed base
Avoid or limit dairy	41%	Demand extends beyond medical need
Dairy-free ice cream market	$2.65B	Material revenue pool
Category growth (CAGR)	15.3%	Fast-expanding, early enough to win
DQ store closures	42	Brand under pressure
DQ average unit volume	about $1.165M (flat)	Needs a growth catalyst

A dairy-free line is therefore simultaneously a **defensive** response to a declining footprint and an **offensive** entry into a high-growth segment.

2. Methodology

Two studies were fielded and analyzed independently, then triangulated. Both used human respondents augmented with AI-generated personas to expand sample coverage, a limitation discussed in Section 7.

2.1 Perceptual Mapping Study

Parameter	Detail
Sample	**n = 52** (32 human plus 20 AI-augmented personas)
Brands	Dairy Queen, Jeni's, Ben & Jerry's, Baskin-Robbins, Andy's Frozen Custard
Attributes	8 (affordability, easy to find, flavor variety, tastes good, health-conscious options, guilt-free, customer-brand relationship, brand preference)
Scale	7-point Likert
Technique	Factor analysis to **2 components, 96.8% of variance explained**

The two extracted dimensions are interpreted as a horizontal **awareness and consideration axis** ("Who?" to "Obsessed") and a vertical **momentum axis** ("Melting Away" to "Scooped Up").

2.2 Conjoint Analysis Study

Parameter	Detail
Sample	**n = 50** (5 human plus 45 AI-augmented personas)
Features	7 (Price, Availability, Type of Ice Cream, Flavor Profile, Milk Type, Certification, Social Cause)
Levels	3 per feature
Design	**L18 orthogonal array, 18 product profiles**
Scale	7-point purchase-likelihood
Estimation	OLS regression to part-worth utility values

3. Perceptual Mapping: Where DQ Stands

Finding: DQ is under-differentiated and skews toward low consideration. It clusters near the origin with Jeni's, while Ben & Jerry's anchors the high-affinity "Obsessed" pole and the smiley-face zone (Scooped Up) sits unoccupied by DQ.

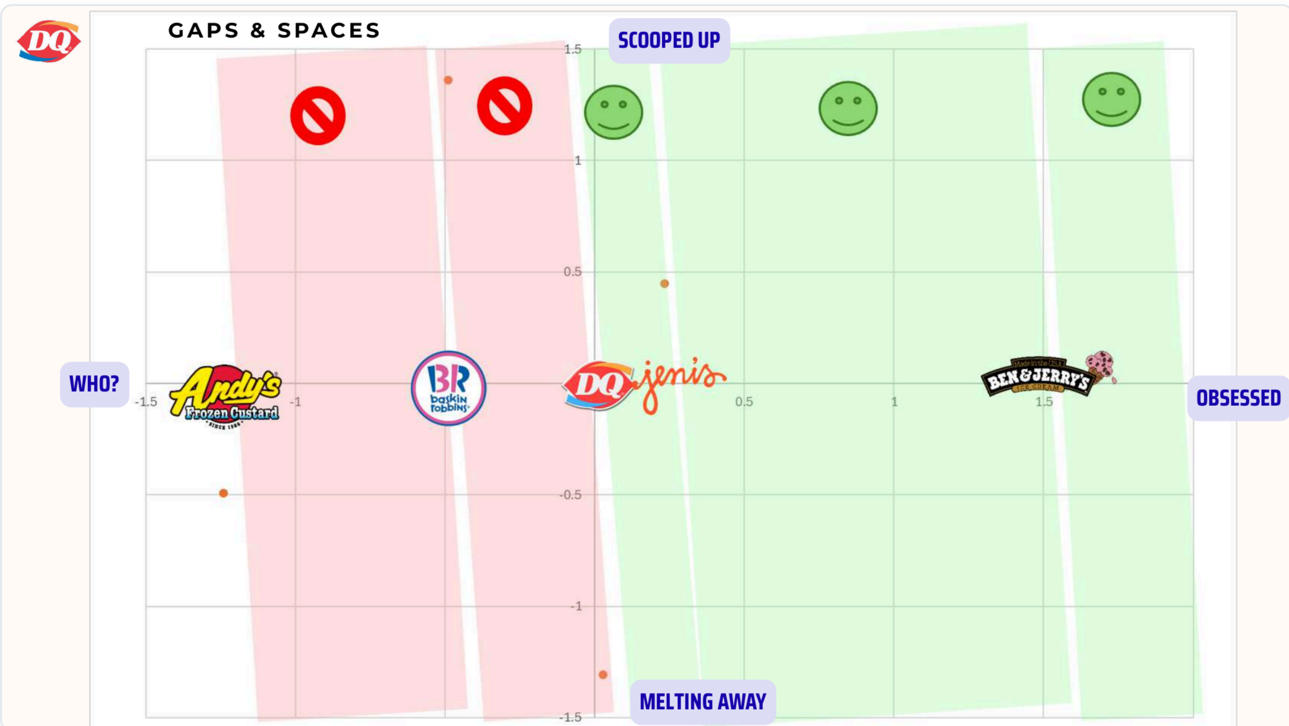


Figure 1. Brand positioning across the two factor dimensions. Green marks the high-affinity "scooped up" zone; red marks the low-consideration "who?" zone. DQ sits at the boundary, near Jeni's, well short of Ben & Jerry's.

The attribute-vector view explains *why*. The longest, most influential vectors, "easy to find," "tastes good," "flavor varieties," and "health-conscious options," point toward the right ("Obsessed") quadrant where DQ is **not** strongly placed. DQ's relative strength is "affordable" and distribution reach, not perceived taste leadership or guilt-free credentials.

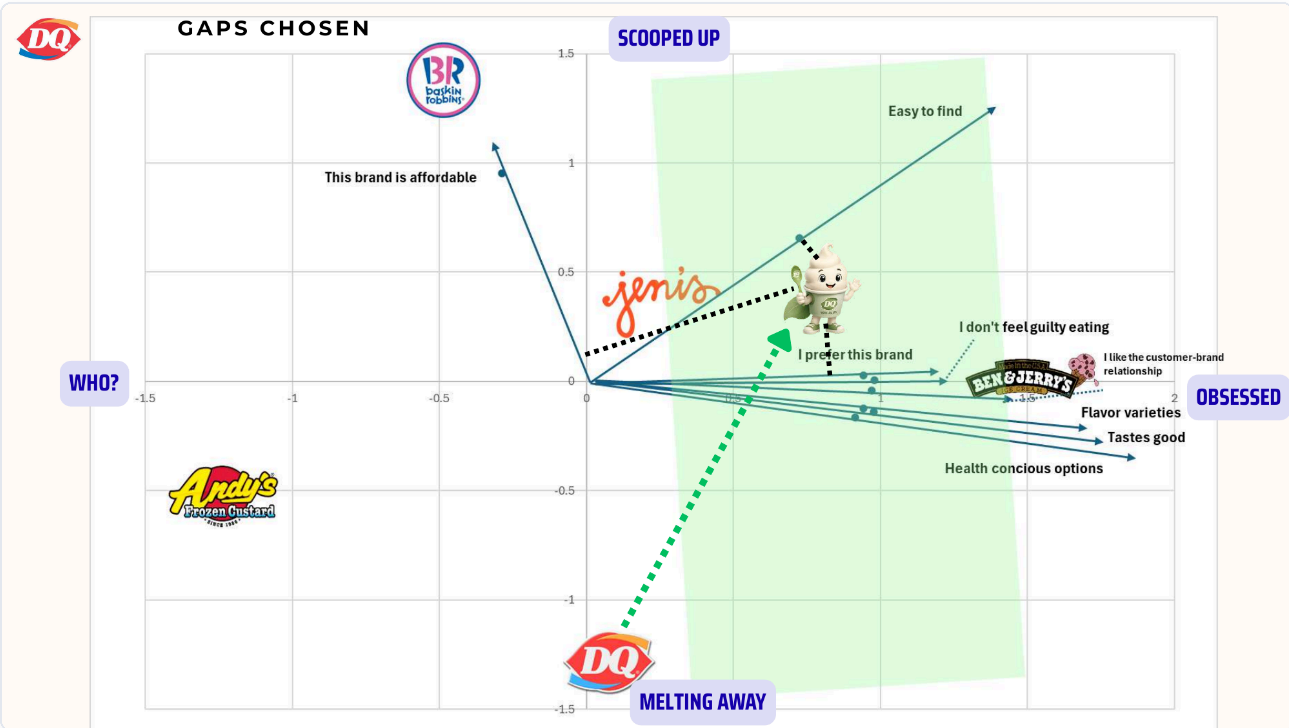


Figure 2. Attribute vectors. "Easy to find" and "tastes good" are the dominant pull directions, both addressable by DQ through its store footprint and an indulgent dairy-free product.

Strategic read: DQ's gap is not awareness of the brand, it is the absence of a *reason to choose DQ for dairy-free*. The map says the open lane is **"affordable, easy to find, tastes good,"** which is precisely the position a $4.99 indulgent dairy-free Blizzard at 4,200+ locations would occupy.

4. Conjoint Analysis: What Actually Drives Purchase

4.1 Feature Importance

Finding: Price, Availability, and Type of Ice Cream account for about 62% of purchase decisions. Certification and social cause are nearly irrelevant.

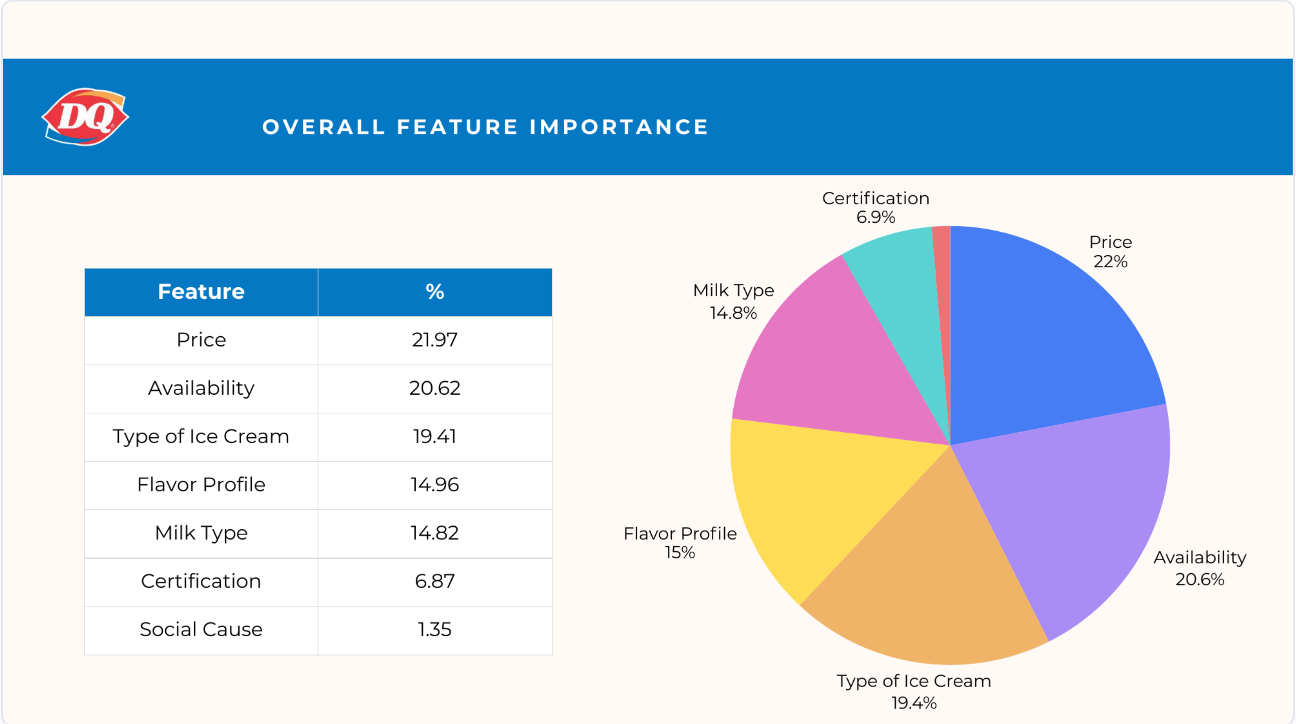


Figure 3. Relative feature importance from the conjoint model.

Feature	Importance
Price	**21.97%**
Availability	20.62%
Type of Ice Cream	19.41%
Flavor Profile	14.96%
Milk Type	14.82%
Certification	6.87%
Social Cause	1.35%

The implication is blunt: **marketing budget spent on vegan or kosher certification badges, or on cause marketing, will not move the needle.** Spend on price, distribution, and product format.

4.2 Part-Worth Utilities

Finding: The optimal level of each feature is clear and consistent. Lower price, indulgent flavor, oat-milk base, and walk-in availability each carry the highest utility within their feature.

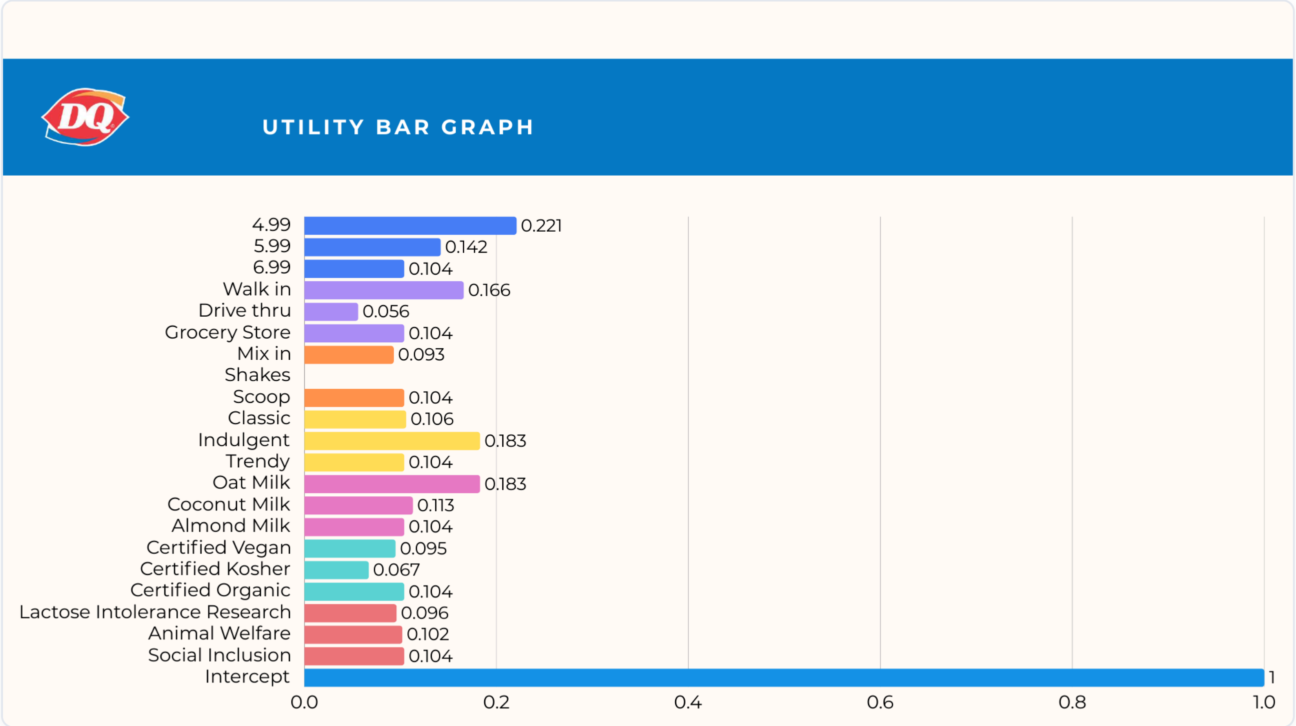


Figure 4. Part-worth utilities by level. Highlighted winners: $4.99 (0.221), Walk-in (0.166), Indulgent (0.183), Oat Milk (0.183).

Lever	Winning level	Utility
Price	$4.99	**0.221**
Milk type	Oat milk	**0.183**
Flavor	Indulgent	**0.183**
Availability	Walk-in	**0.166**
Coconut milk (alt)	secondary	0.113
Format	Scoop	0.104

4.3 Modeled Total Preference Utility

Finding: A correctly configured DQ beats every competitor. Summing the optimal part-worths, DQ's total preference utility reaches **1.065**, the highest in the set.

COMPETITOR TOTAL PREFERENCE UTILITY

BEN & JERRY'S	
Utility	**TPU**
$6.99	0.104
Grocery Store	0.104
Scoops	0.093
Indulgent	0.183
Coconut Milk	0.113
Certified Vegan	0.095
Social Inclusion	0.104
TOTAL	**0.796**

Jenni's	
Utility	**TPU**
$6.99	0.104
Walk in	0.166
Scoops	0.093
Trendy	0.104
Oat Milk	0.183
Certified Organic	0.104
Animal Welfare	0.102
	0.856

Baskin Robbins	
Utility	**TPU**
$5.99	0.142
Drive Thru	0.056
Scoops	0.093
Classic	0.106
Oat Milk	0.183
Certified Kosher	0.067
Lac. Int. Research	0.096
	0.743

Andy's Custard	
Utility	**TPU**
$4.99	0.221
Drive thru	0.056
Mix-ins	0.093
Indulgent	0.183
Almond Milk	0.104
Certified Kosher	0.067
Social Inclusion	0.104
	0.828

Dairy Queen	
Utility	**TPU**
$4.99	0.221
Walk in	0.166
Scoops	0.093
Indulgent	0.183
Oat Milk	0.183
Certified Organic	0.104
Social Inclusion	0.104
TOTAL	**1.054**

Figure 5. Total preference utility by competitor, configured as observed or optimal.

Brand	Total Preference Utility
Dairy Queen (optimal)	**1.065**
Jeni's	0.856
Andy's Frozen Custard	0.828
Ben & Jerry's	0.796
Baskin-Robbins	0.743

DQ TOTAL PREFERENCE UTILITY

Utility	Best TPU
$4.99	0.221
Walk in	0.166
Scoops	0.104
Indulgent	0.183
Oat Milk	0.183
Certified Organic	0.104
Social Inclusion	0.104
TOTAL	**1.065**

Utility	Optimal TPU
$4.99	0.221
Walk in	0.166
Mix-in	0.093
Indulgent	0.183
Oat Milk	0.183
Certified Organic	0.104
Social Inclusion	0.104
	1.054

Utility	Worst TPU
$6.99	0.104
Drive thru	0.056
Shakes	0
Trendy	0.104
Almond Milk	0.104
Certified Kosher	0.067
Lac. Int. Research	0.096
	0.531

Figure 6. DQ utility under Best (1.065), Optimal (1.054), and Worst (0.531) configurations. The spread shows how much value is destroyed by the wrong build, most of it from mispricing ($6.99) and the wrong format (shakes and drive-thru).

The distance between the Best (1.065) and Worst (0.531) DQ builds, a 2x swing, is the core message: the opportunity is real but only if the SKU is configured to the data.

5. Survey Findings: Demand Is Real and Price-Sensitive

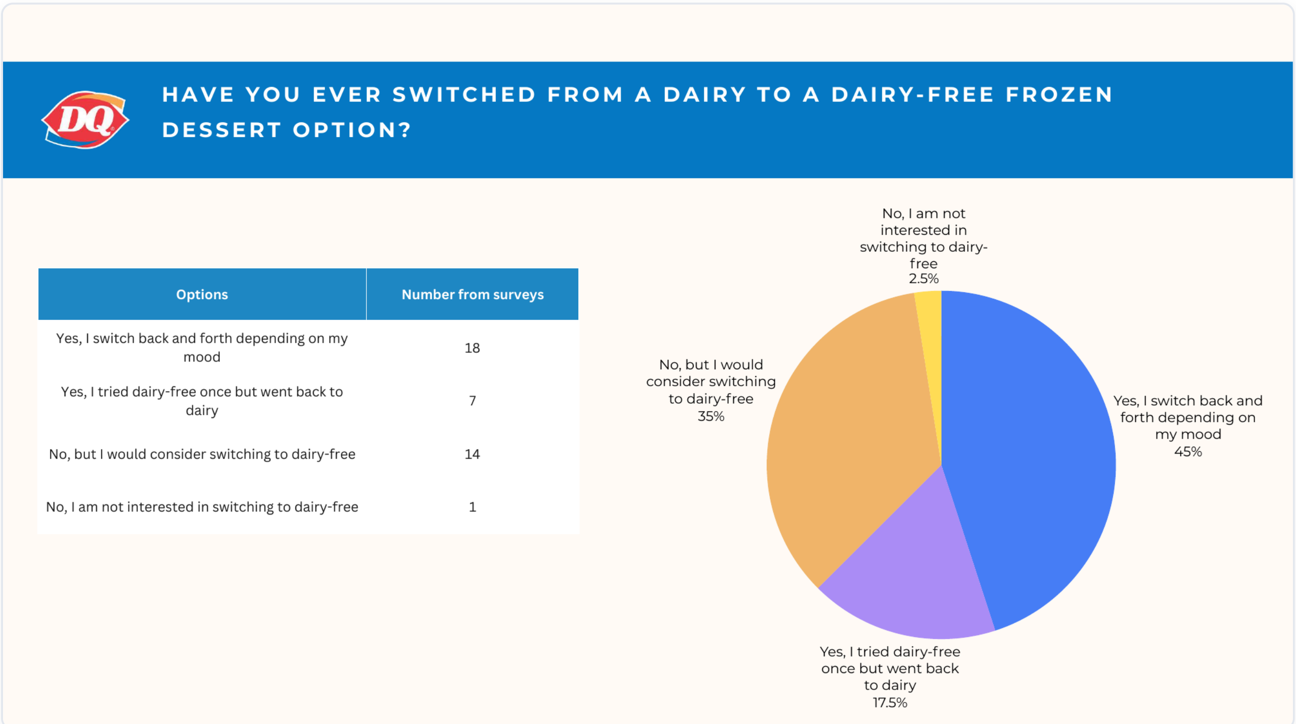
The perceptual survey also captured behavior and intent that frame the launch.

Buyers are willing, but only at the right price. Roughly 55% would pay the same as regular ice cream or just $1 more for dairy-free. Premium tolerance drops sharply beyond +$2.

Options	Number from surveys
Same as regular ice cream	19
$1 more	12
$2 more	8
$3+	1

Figure 7. Premium willingness for dairy-free. The mass sits at "same" and "+$1," validating the $4.99 entry price.

Switching intent is high. 80% have switched, switch back and forth, or would consider switching to dairy-free; only 2.5% are hard "not interested."



Options	Number from surveys
Yes, I switch back and forth depending on my mood	18
Yes, I tried dairy-free once but went back to dairy	7
No, but I would consider switching to dairy-free	14
No, I am not interested in switching to dairy-free	1

Figure 8. Dairy-to-dairy-free switching behavior.

Distribution matters. Specialty shops (45%) and grocery (42.5%) dominate today; quick-service restaurants take only 10%, a white space DQ's 4,200+ locations can attack directly.

WHAT IS YOUR PREFERRED MODE TO PURCHASE ICE CREAM?

Options	Number from surveys
Quick-service restaurant	4
Specialty ice-cream shops	18
Grocery stores	17
Convenience stores	1

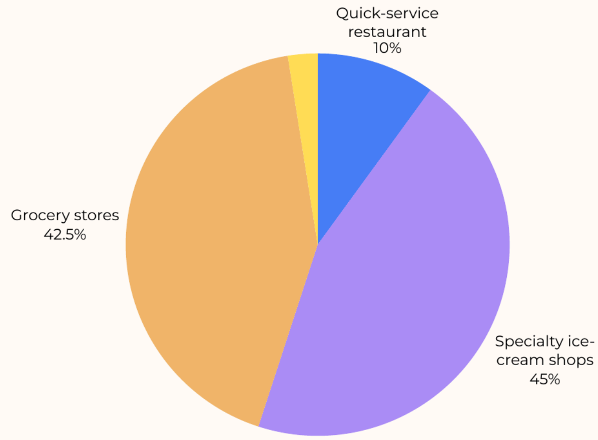


Figure 9. Preferred purchase channel. QSR is under-served, DQ's structural advantage.

Flavor beats values in the decision. 52.5% decide based on flavor availability; only 5% on brand values, reinforcing the conjoint's "lead with indulgence" signal.

HOW DO YOU TYPICALLY DECIDE WHICH DAIRY-FREE FROZEN DESSERT BRAND TO PURCHASE?

Options	Number from surveys
I always buy the same brand out of habit	8
I compare options and choose based on price	8
I choose based on flavor availability	21
I choose based on brand values	2
I try different brands each time	1

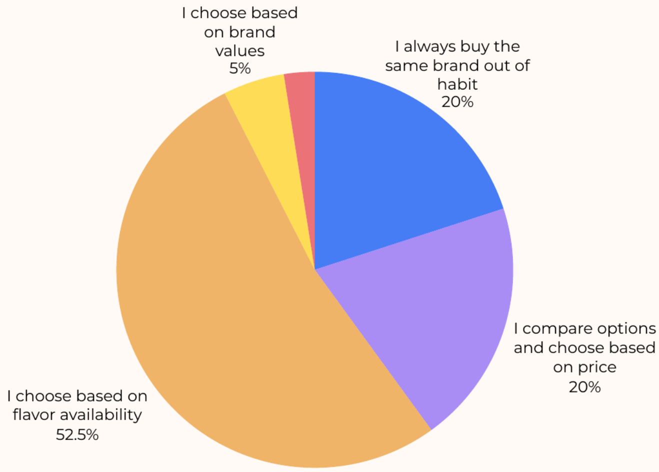


Figure 10. How buyers choose a dairy-free brand.

Sample skew (a caveat). 80% of respondents were Midwest-based, favorable to DQ's footprint but a geographic concentration to flag when generalizing.

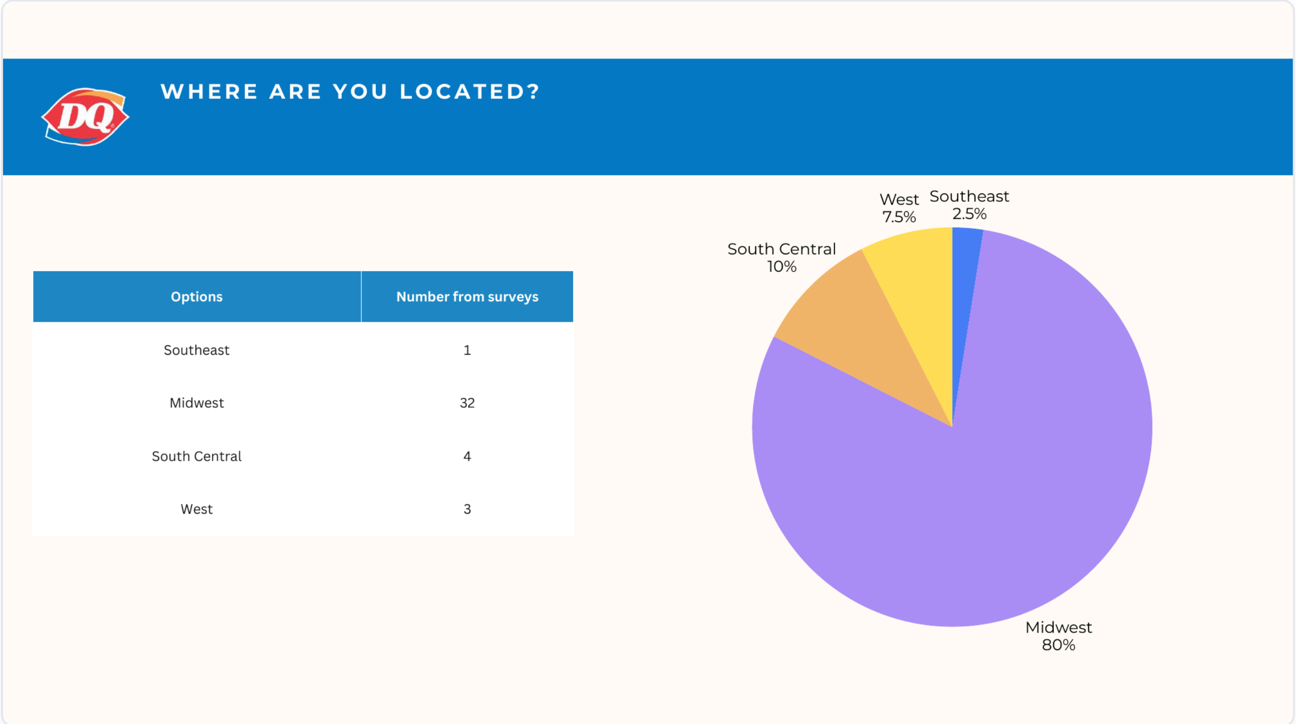


Figure 11. Respondent location distribution.

6. Operational Challenges and Growth Math

A plant-based base is not free. The model and category benchmarks indicate:

Lever	Impact
Plant-based base food cost	**+30% to +55%** vs. dairy
Recommended premium pricing	**+$1.00 to +$1.75** per unit
Expected ticket value lift	**+3% to +5%**
Expected dessert category lift	**+4% to +7%**

The arithmetic works because the premium tolerance (Section 5) and the high feature-importance of price (Section 4.1) line up. A **$4.99 entry with a modest premium variant** captures price-sensitive switchers while protecting margin on a higher-cost base. The 4,200+ walk-in network means incremental dessert attach has near-zero distribution cost.

7. Recommendations

Ranked by leverage and supported directly by the two studies.

1. **Launch a dairy-free Blizzard at $4.99 on an oat-milk base.** This is the literal output of the conjoint: top price utility (0.221), top milk utility (0.183), top flavor utility via indulgent (0.183). *Primary move.*

2. **Lead with indulgence, not certification.** Flavor drives 52.5% of decisions and indulgent carries the highest flavor utility; certification and social-cause messaging is statistically negligible (6.87% and 1.35%). Spend creative on taste, not badges.
3. **Sell it where DQ already wins, the walk-in.** Walk-in carries +0.166 utility and QSR is the under-served channel (10% today). DQ's 4,200+ locations turn distribution into a moat.
4. **Roll out an oat-milk scoop line across the full walk-in network** as the scalable follow-on, using coconut milk as the secondary variant (0.113) for flavor breadth.

Supporting partnerships (optional, not core): co-brand mix-ins with **UNREAL** and source oat base from **Minor Figures** to accelerate credibility and supply. Useful, but the data shows the product configuration matters far more than the partner logos.

8. Conclusion

Both methods converge on one answer. Perceptual mapping shows DQ has an open lane, "affordable, easy to find, tastes good," that no competitor owns. Conjoint analysis specifies exactly how to occupy it: **$4.99, indulgent, oat-milk, walk-in.** Configured this way, DQ's modeled preference utility (1.065) leads the category. The market is large ($2.65B), fast-growing (15.3% CAGR), and DQ's declining footprint needs precisely this kind of catalyst. The recommendation is not "consider dairy-free," it is "launch the specific SKU the data describes."

Appendix: Data and Limitations

A. Category data sources. Lactose intolerance about 36% of U.S. population; 41% avoid or limit dairy; dairy-free ice cream market $2.65B at 15.3% CAGR; DQ 42 closures, about $1.165M flat AUV. Compiled from industry and category reports referenced in the source deck.

B. Perceptual mapping. n = 52 (32 human plus 20 AI personas); 5 brands across 8 attributes; 7-point Likert; factor analysis, 2 components, 96.8% variance.

C. Conjoint analysis. n = 50 (5 human plus 45 AI personas); 7 features across 3 levels; L18 orthogonal design (18 profiles); 7-point purchase-likelihood; OLS estimation.

D. Market share (modeled). DQ 24.6%, Jeni's 20%, Andy's 19.4%, Ben & Jerry's 18.6%, Baskin-Robbins 17.4%.

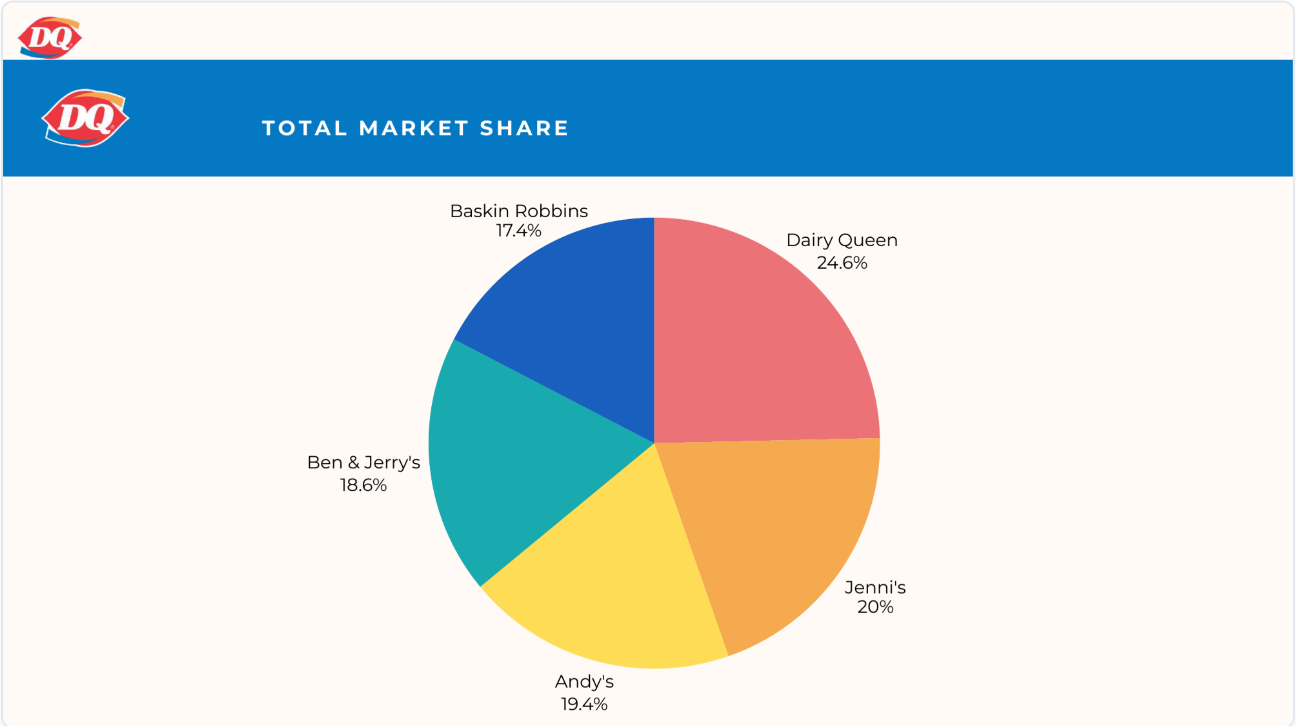


Figure A1. Modeled market share derived from total preference utilities.

E. Limitations.

- **AI-augmented samples.** Both studies blend human respondents with AI personas (38% of perceptual, 90% of conjoint). Utility magnitudes should be read as directional, not population-precise.
- **Geographic skew.** 80% Midwest respondents inflate DQ-favorable signal; national generalization is limited.
- **Small human n in conjoint** (5) means the orthogonal design and AI personas carry most of the estimation weight.
- **Stated vs. revealed preference.** Purchase-likelihood scales overstate real conversion; treat utility rankings as relative, not absolute demand forecasts.